

EDUCATION

University of Pennsylvania, *Ph.D. in Computer and Information Science*

September 2022 – Present

Harvard University, *M.E. in Computational Science and Engineering*

September 2019 – May 2021

Xiamen University, *B.E. in Material Science and Engineering*

September 2015 – June 2019

INDUSTRY EXPERIENCE

Adobe, Evaluation and Experience Lab

May 2026 – August 2026

*Research Scientist Intern**San Jose, CA*

- Led architecture and implementation of a **review-and-refine agent** for Adobe's **multi-agent audience-generation system**, using agent handoffs to audit generated queries for campaign requirements, schema/data evidence, and upstream reasoning.
- Designed a checklist-driven **evaluation framework** and **tool-calling harness** with context engineering, decomposing queries into reviewable predicates and retrieving schema, metadata, data distributions, query results, and agent provenance on demand; parallel execution cut review latency **42%**.
- Engineered an **execution-grounded refinement loop** with deterministic validation, compositional revalidation, retries, and partial recovery across **1,850 predicate-level evaluations**; recommendations resolved **93.2% of review issues** and cut average human review from **15 to 3 min**.

Microsoft Research, PROSE Team

April 2025 – July 2025

*Research Intern**Remote*

- Architected InvisibleMentor (CHI '26), a **multimodal VLM pipeline** that reconstructed software workflows from screen recordings without application logs or instrumentation, achieving **0.91 F1** on a 350-instance benchmark.
- Designed **stateful long-context inference** with hierarchical temporal segmentation and parallel GPT-4.1 VLM calls, reducing one-hour video analysis latency from **6 to 2 minutes**.
- Built an **execution-based evaluation framework** for generated recommendations; **58/60 (96.7%)** completed the intended workflow without repair when replayed in target applications.
- Evaluated recommendation quality in a controlled study, reducing users' workflow interactions by **32.4%** on average while preserving task outcomes.

SELECTED RESEARCH

Execution Digests (CHI '27 submission) Agent Harness, Agent Evals

January 2026 – August 2026

- Built a **20-task benchmark dataset** from **18 public Python repositories**, with reproducible revisions and **evaluation rubrics** defining expected verdicts, required evidence, and sufficiency criteria.
- Architected a **coding-agent evaluation framework** around Claude Code Sonnet with **15 typed tools** over indexed runtime traces, enabling agents to perform targeted real-world software-engineering investigations across **50K+ execution events** without full-trace context loading.
- Built a deterministic **grader** that automatically verified verdicts, source anchors, and deciding runtime values, achieving **100% deciding-evidence coverage vs. 65%** for unrestricted Claude Code at equal verdict accuracy.
- Engineered **agent orchestration** with deterministic-first execution, model routing, caching, and checkpoint recovery, cutting time to a correct review **45%** and raising 15-minute completion from **50% to 95%**.

TrailBlazer (UIST '25) LLM Workflows, Agent Orchestration

January 2024 – April 2025

- Architected a four-stage **LLM-guided code-exploration workflow** with model routing, schema-constrained outputs, and LSP/compiler tools for grounded, machine-verifiable exploration.
- Parallelized candidate scoring and exploration over an explicit **agent search graph**, cutting candidate-analysis latency **4x** through batched concurrent model calls.
- Compiled minimal evidence walkthroughs from agent exploration paths; evaluation reduced task time **34%** with **100% completion vs. 65%**.

Ivie (CHI '24) AI-Assisted Coding, Developer Tools

October 2022 – December 2023

- Built an **AI-assisted coding tool** in VS Code with parallel multi-granularity prompting, structured generation, and token-streamed parsing, achieving **97% valid explanations** in a 100-output generation audit.
- Integrated anchored, just-in-time explanations into the editor and evaluated them with eye-tracking, raising code-comprehension accuracy from **65.0% to 90.2%**.

ExampleNet (CHI '21) Best Paper and Honorable Mention, Program Analysis

September 2019 – May 2021

- Built a **three-pass interprocedural analyzer** normalizing 63 TensorFlow/Keras APIs into 27 canonical layer types, recovering **1,377 layers across 62 repositories**.
- Adapted **multiple-sequence alignment** for cross-project neural-architecture comparison and pattern discovery.

TECHNICAL SKILLS

Agent Systems: Coding agents, AI-assisted coding, agent harnesses, tool/function calling, multi-agent orchestration**Evaluation:** Evaluation frameworks, rubric design, benchmark design, deterministic graders, agent evals**LLM/Multimodal:** LLMs, VLMs, multimodal reasoning, structured outputs, long-context inference, context engineering, embeddings, reranking, RAG, MCP**ML/Engineering:** PyTorch, TensorFlow/Keras, Python, TypeScript/JavaScript, SQL, React, FastAPI, Node.js, Docker, static/dynamic analysis, AST/LSP